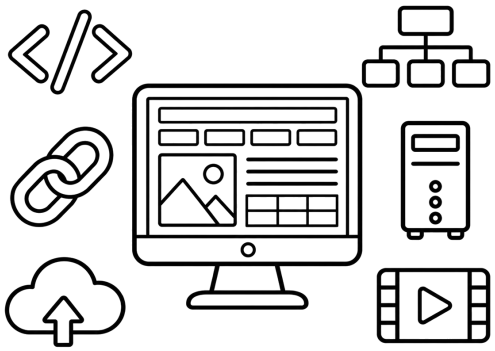
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 11



UNIT 5
Web Designing Using Multimedia

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Name the kinds of web applications and plan a website for its users.
- [ ] Organise web content with a site map, menus and hyperlinks.
- [ ] Choose media files of a small size and use them with copyright.
- [ ] Write an HTML document in Notepad and open it in a web browser.
- [ ] Format text, colours, images, lists, links and tables with HTML tags.
- [ ] Compare web authoring tools and CMS, static and dynamic websites.
- [ ] Publish a website with a web host, a domain name and FTP; maintain it.

HOW TO USE THIS BOOK

Read each chapter with your teacher, do the practical tasks and then write the exercises. The **WRITING SPACES** are for the notes your teacher gives you in class - write them neatly: they are your revision notes for the O/L examination.



THE WORDS YOU WILL LEARN

Web application	User group	Web content	Site map
Main menu	Navigation	Copyright	Fair use
HTML	Tag	Element	Attribute
Value	Empty tag	Hyperlink	Table
WYSIWYG	Web authoring tool	CMS	Static website
Dynamic website	Web host	Domain name	FTP

"A good website starts with a good plan. Think of the users first, then write the code."

- THE KING OF ICT

WRITING SPACE 1 - What this unit is about

write what your teacher tells you

CHAPTER 1

Planning a Website

? A WEBSITE FOR OUR SCHOOL

"Our ICT group must design the school website. Where do we start?" "With a plan!"

Think of its purpose, its users, its content and how to maintain it (textbook pages 125 - 134).



5.1 Web applications

Web application	These websites ...	Examples (textbook 5.1.1)
Information communication	educational and health information	www.nie.lk, www.surgery.lk, www.webopedia.org
e Commerce	sell goods and services on the Internet	www.farlin.lk, www.lego.lk, www.amazon.com
Entertainment	songs, films, teledramas, games ...	www.islandcricket.lk, www.netflix.com, www.youtube.com
Advertising	goods and services of organizations	www.onclickads.net, www.adcash.com, www.myadvertisingpays.com
News	publish news using multimedia	www.itnnews.lk, www.rupavahini.lk, www.bbc.co.uk
Social media	social networking among groups	www.facebook.com, www.linkedin.com, www.twitter.com
Search engine	find information with a word or phrase	www.google.com, www.msn.com, www.bing.com
Personal	popular persons share their information	www.president.gov.lk, www.arthurcclarke.net, www.nelsonmandela.org

5.2 Web users and web content

Those who access and use a website are its **users**; in designing a website the user is important (5.1.2). www.srilanka.lk can be used in all the three languages; www.trc.gov.lk lets users change the font size and reach social networks (Figures 5.1, 5.2).

i THINK OF THEIR

- Language and culture
- Age
- Educational level
- Gender
- Preferences

? HOW TO STUDY THEM

- Conducting a study on the target users.
- Analysing similar websites.

Web content is all the audio-visual information and services a user comes across in a website; select it for the needs of the users (5.1.3): texts, images (photos, clip arts, drawings, scans), sound, video, animations, copyrights and update information, hyperlinks, advertisements, search facilities and facilities to download software.

"Plan the pages on paper. The best websites are designed before they are built."

- THE KING OF ICT

5.3 Content management

Content management is the systematic organisation of the selected content of a website, so that users can use it more effectively (5.1.4). A well organised website is easy to use, users find the information they need fast, it attracts users, it is easy to update and it has room to add new features later.

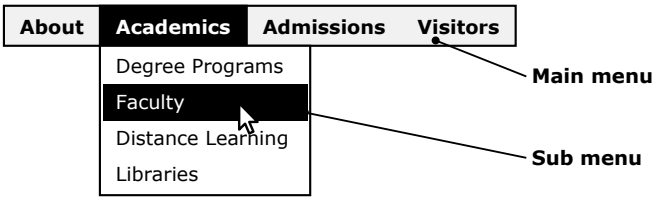


Figure 5.4 - a main menu and a sub menu

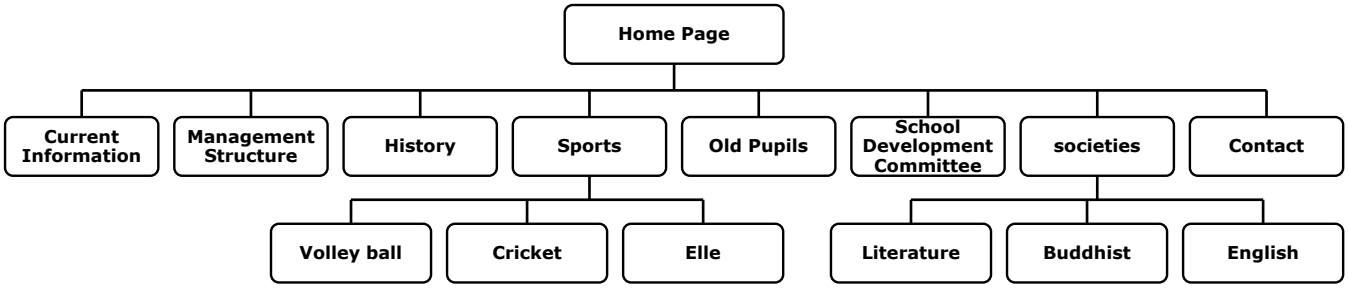


Figure 5.3 - the site map of a school website: the boxes are web pages, the lines are hyperlinks

Method	How it helps
1. Site map	the web pages (boxes) and the hyperlinks among them (lines).
2. Summary on the home page	a main menu with sub menus, and hyperlinks, save space.
3. Information in brief	limit the content of a single web page.
4. Lists and indentation	numbered and bulleted lists: no long, tiring paragraphs.
5. Easy navigation	hyperlinks from page to page and back to the home page.
6. Updating	easy to update; copyright and updated time on each page.

WRITING SPACE 2 - Content management — write what your teacher tells you

5.4 Structure and layout

Backgrounds, tables, frames, font types, sizes and shapes (5.1.5):

1. **Keep the identity of the website on each page** - the same backgrounds, font types, font sizes and font shapes.
2. **Highlight important information** - with other font types, font sizes or colours.
3. **Access from any computer** - screens differ in size and resolution: the web page should adjust its size to the screen automatically.

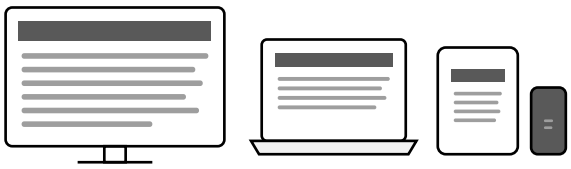


Figure 5.6 - various sizes of screens

5.5 Selecting media resources

Size. Bigger media (texts, pictures, sounds, videos, animations) mean more data to download from the web server to the client computer: users wait long and may leave the site (5.1.6).

Graphics	Video files	Audio files
- Reduce the pixels: remove unnecessary parts with Crop / Trim - Graphic compression	- Reduce the frame size - Reduce the resolution - Adjust the frame rate - Remove unnecessary frames and sound	- Reduce the bit rate - Reduce the sample rate - Reduce the number of channels - Remove or reduce unnecessary sound

Copyright. Get the owner's permission to use media made by another person; if not, it is unethical and may lead to legal issues. By **fair use**, media can be used without permission for news reporting, academic purposes and criticism.

WRITING SPACE 3 - The size of media files — write what your teacher tells you

DO IT - PRACTICAL 1 (ACTIVITIES, PAGES 132, 134) - PLAN A WEBSITE

Websites to examine: **www.srilanka.lk, www.schoolnet.lk, www.doenets.lk, www.bbc.co.uk/education**.

1	With your teacher, look at the users, content, menus and layout of each website.
2	Write one strength and one weakness in the design of each website.
3	In your group, plan a school website: its purpose, its users and its content.
4	Draw its site map. Choose one background and one font style for all the pages.
5	Plan small media files; ask the owners before you use their media.

A True or false

8 marks

Write T (true) or F (false) in the box.

1.		Web content includes all the information and services a website gives its users.
2.		Age, gender and educational level help us to understand the users of a website.
3.		Content management is the systematic organisation of the selected content of a website.
4.		One web page should hold as much information as possible.
5.		A site map shows the web pages to be created and the hyperlinks among them.
6.		Each page of one website should have a different background and different fonts.
7.		A website should adjust the size of its web page to the screen automatically.
8.		You may use any picture of another website without permission.

B Match the websites

8 marks

Write the letter of the kind of web application.

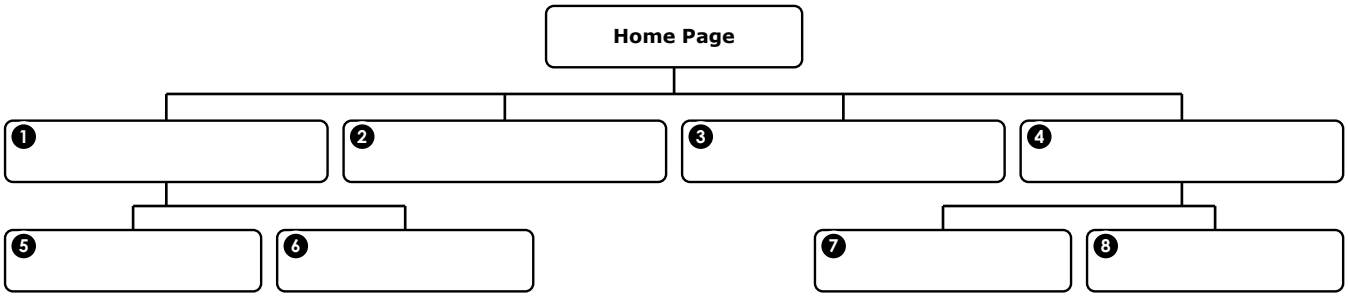
A - Website		B - Web application
1. www.nie.lk		**A.** Search engine
2. www.amazon.com		**B.** e Commerce
3. www.youtube.com		**C.** Personal
4. www.adcash.com		**D.** News
5. www.rupavahini.lk		**E.** Information communication
6. www.linkedin.com		**F.** Social media
7. www.bing.com		**G.** Advertising
8. www.nelsonmandela.org		**H.** Entertainment

C Complete the site map

5 marks

Write the page names in the empty boxes (half a mark each).

The Science Society website has a Home Page with hyperlinks to four pages: **About Us**, **Events**, **Gallery** and **Contact**. The Events page links to **Science Day** and **Quiz**. The Gallery page links to **Photos** and **Videos**.



9. How many web pages must be created for this website? (1 mark) .....

D Make the media smaller

8 marks

Tick the kind of file each way is used for.

A way to reduce the file size	Graphics	Video	Audio
Reduce the bit rate	☐	☐	☐
Remove unnecessary parts with Crop / Trim	☐	☐	☐
Reduce the frame size	☐	☐	☐
Reduce the number of channels	☐	☐	☐
Graphic compression	☐	☐	☐
Adjust the frame rate	☐	☐	☐
Reduce the sample rate	☐	☐	☐
Remove unnecessary frames and sound	☐	☐	☐

CHAPTER 2

HTML: Your First Web Page

? WHERE IS MY WEB PAGE?

"My page opens in Notepad again, not in the browser!" "Save it as .html with All Files!"

Write HTML, save it as a web page and format its title text (textbook pages 134 - 144).



5.6 HTML and your first web page

Hyper **T**ext **M**arkup **L**anguage (**HTML**) is the language used for documents in the World Wide Web. An HTML document that a web browser can display is a **web page**. Unlike Pascal, HTML codes show the web browser how the content should be displayed, so HTML is not a computer (programming) language but a **markup language** (5.1.7). Instead of Notepad, **HTML editors** can be used: **CoffeeCup** (www.coffeecup.com/html-editor), **Seamonkey Composer** (www.seamonkey-project.org), **Eclipse** (marketplace.eclipse.org/content/html-editor-wtp).

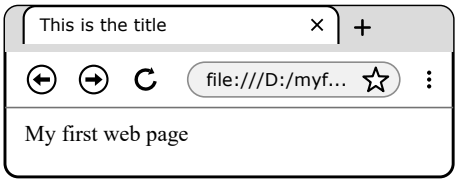


Figure 5.10 - myfirst.html in a browser

> DO IT - PRACTICAL 2 (ACTIVITY NO 01, STEPS 1 - 8) - MY FIRST WEB PAGE

1	Open Notepad: **Start** → **All Programs** → **Accessories** → **Notepad**.
2	Type the HTML statements of Figure 5.8.
3	**File** → **Save As**: **myfirst.html**, Save as type **All Files**.
4	Open myfirst.html: the web page shows in the browser.
5	Write other phrases and create more web pages.
6	Install an HTML editor: CoffeeCup, Seamonkey or Eclipse.
7	CoffeeCup: **File** → **New HTML document**; insert your phrases.
8	Save it as **coffeecup.html**; use the **preview tool**.

Untitled - Notepad

```
<html>
   <head>
      <title> This is the title </title>
  </head>
      <body>
          My first web page
      </body>
</html>
```

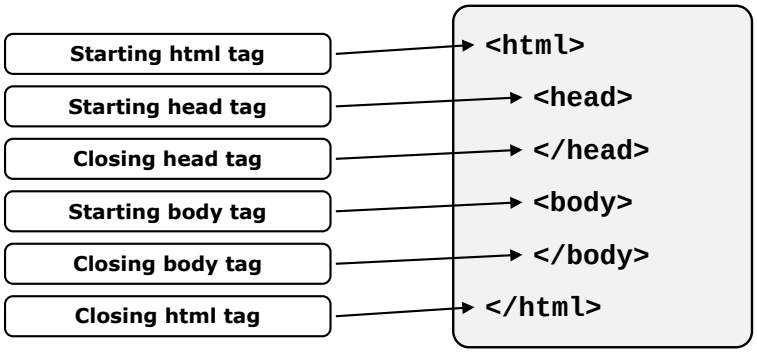
Figure 5.8 - HTML in Notepad

"Every tag you open, close it. Neat code makes happy browsers."

- THE KING OF ICT

WRITING SPACE 4 - HTML write what your teacher tells you

5.7 The basic structure of an HTML document

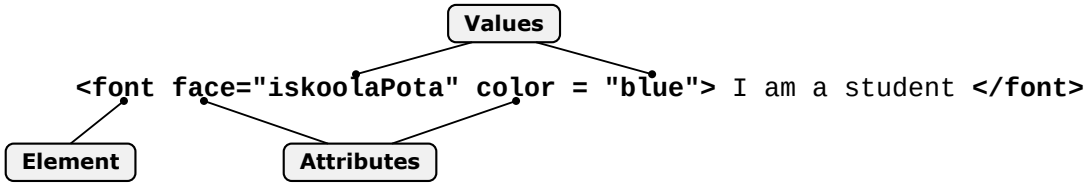


What we write in angular brackets, like **<html>**, **<head>** and **<title>**, are **HTML tags**. Mostly a tag has an opening and a closing tag. The two main parts are the **head** and the **body**. The content goes in the body; the **<title>** is shown on the title pane (5.1.8).

Figure 5.13 - the basic structure of an HTML document

5.8 Features of HTML

1) HTML tags (HTML codes) show the web browser how the web page should be displayed.
2) A tag begins and ends with angular brackets **< >**: ** ** begins a paragraph.
3) Closing tags are given in pairs, with a slash **/** and the name: ** ** ... ** **.
4) Tags are not case sensitive: **<TITLE>**, **<Title>** and **<title>** are the same.
5) No error message for a wrong tag: the web browser ignores it.
6) Three parts: **element**, **attribute**, **value**. Attributes go in the opening tag only; values within inverted commas.



The three parts of a tag (textbook page 138)

WRITING SPACE 5 - Element, attribute, value write what your teacher tells you

"Angular brackets open the door; the slash closes it."

- THE KING OF ICT

5.9 The title text: header, font and center tags

Let us design the web page of Figure 5.14, a short introduction about Sri Lanka (5.1.10):

```
<html>
    <head>
        <title> Sri Lanka </title>
    </head>
    <body>
        <h2><center><font face="Arial" color="blue">
            SRI LANKA
            </font></center></h2>
    </body>
</html>
```

Figure 5.15 - the title text

Header tags **<h1>** to **<h6>**: **<h1>** makes the biggest and boldest letters; they get smaller down to **<h6>**.

The font tag has three main attributes: **size** (1 to 7; the default is 3), **face** (the type of letters, e.g. "Arial") and **color**.

** ** aligns a title or a picture to the centre.



The output

5.10 Colours

A value for **color**: (i) the name of one of the **sixteen standard colours**: Black, Gray, Silver, White, Yellow, Lime, Aqua, Fuchsia, Red, Green, Blue, Purple, Maroon, Olive, Navy, Teal (Figure 5.16); or (ii) an **RGB hexadecimal code**: # and six digits from 0 to F - the first two for red, the next two for green and the last two for blue, e.g. **#0000FF** blue: no red, no green, all blue (Figure 5.17).

```
<font size="5" face="verdana" color="fuchsia">SRI LANKA</font>      ← a colour name
<font size="5" face="verdana" color="#0000FF">SRI LANKA</font>      ← an RGB code
```

5.11 Text formatting tags

Like ** **, these tags design texts (Table 5.1): ** SRI LANKA ** shows **SRI LANKA**. More: www.w3schools.com/html, www.tutorialspoint.com/html.

Tag	Letters	Example	Tag	Letters	Example
****	Bold	**SRI LANKA**	**<s>**	Strike out	~~SRI LANKA~~
<i>	Italic	*SRI LANKA*	**<sup>**	Superscript	SRI LANKA
<u>	Underline	SRI LANKA	**<sub>**	Subscript	SRI LANKA
****	Emphasis	*SRI LANKA*	**<marquee>**	Horizontally moving text	*check it on a page*

WRITING SPACE 6 - Head and body — write what your teacher tells you

DO IT - PRACTICAL 3 (ACTIVITIES, PAGES 142, 144) - TEXT FORMATTING

1	Type the code of Figure 5.18. Write the name of your school in Sinhala in each line (face "iskoola pota").
2	Save As with **Encoding: Unicode** and **Save as type: All Files** (Figure 5.19). Open the page.
3	Compare it with Figure 5.20: the letters get smaller from **<h1>** to **<h6>**.
4	Design it again with other text formatting tags instead of the header tags. Try **<marquee>**.
5	Find more text formatting tags in websites or books; make a table like Table 5.1.

```
<html>
<head><title> Text Formatting </title></head>
<body>
  <font face = "iskoola pota" color = "Green" ><h1> 1. name of your school </h1></font>
  <font face = "iskoola pota" color = "lime" ><h2> 2. name of your school </h2></font>
  <font face = "iskoola pota" color = "olive" ><h3> 3. name of your school </h3></font>
  <font face = "iskoola pota" color = "Navy" ><h4> 4. name of your school </h4></font>
  <font face = "iskoola pota" color = "blue" ><h5> 5. name of your school </h5></font>
  <font face = "iskoola pota" color = "Aqua" ><h6> 6. name of your school </h6></font>
  <font face = "iskoola pota" color = "Teal" ><h6> 7. name of your school </h6></font>
</body>
</html>
```

Figure 5.18 - the HTML document of the activity

A Element, attribute or value?

8 marks

Look at the two tags. Tick one box for each part.

```
<font size = "5" face = "verdana" color = "#0000FF">
<p align = "justify">
```

Part of the tag	Element	Attribute	Value
font	☐	☐	☐
size	☐	☐	☐
"5"	☐	☐	☐
"verdana"	☐	☐	☐
color	☐	☐	☐
"#0000FF"	☐	☐	☐
p	☐	☐	☐
"justify"	☐	☐	☐

B Read the colour codes

6 marks

In #RRGGBB, FF is the most of a colour and 00 is none. Write the colour.

Code	Colour	Code	Colour
#FF0000		#000000	
#FFFFFF		#FFFF00	
#FF00FF		#00FFFF	

C True or false

8 marks

Write T (true) or F (false) in the box.

1.		HTML is a programming language like Pascal.
2.		A web page typed in Notepad is saved as All Files, with a name that ends in .html.
3.		**<TITLE>**, **<Title>** and **<title>** do the same thing.
4.		The web browser shows an error message when a tag is written incorrectly.
5.		In the colour code #0000FF, the last two digits are for blue.
6.		Attributes are written in the closing tag.
7.		The font size can be from 1 to 7; the default is size 3.
8.		**<h6>** makes the biggest and the boldest letters.

D Match the formatting tags

8 marks

Write the correct letter in the box.

A - Tag		B - How the letters look
1. **<b>**		**A.** Strike out
2. **<i>**		**B.** Horizontally moving text
3. **<u>**		**C.** Italic
4. ****		**D.** Subscript
5. **<s>**		**E.** Bold
6. **<sup>**		**F.** Emphasis
7. **<sub>**		**G.** Superscript
8. **<marquee>**		**H.** Underline

E Complete the code

6 marks

It shows SRI LANKA in Arial, blue, centred, as a h2 heading. Two words are extra.

srilanka.html - Notepad

```
<html>
<head>
    <❶_____ > Sri Lanka </title>
</❷_____ >
<body>
    <h2><❸_____ ><font ❹_____ ="Arial" ❺_____ ="blue">
    SRI LANKA
    </font></center></❻_____ >
</body>
</html>
```

WORD BANK

color	h2
size	title
body	center
face	head

"Save, then refresh. That is the secret of every web designer."

- THE KING OF ICT

CHAPTER 3

Images, Lists, Links and Tables

OUR SRI LANKA PAGE

"Our page has its title. Now: a map, a paragraph, a list, a link and a table!"

"Each part has its own tag."

In this chapter you finish the web page of Figure 5.14, step by step (textbook pages 144 - 156).

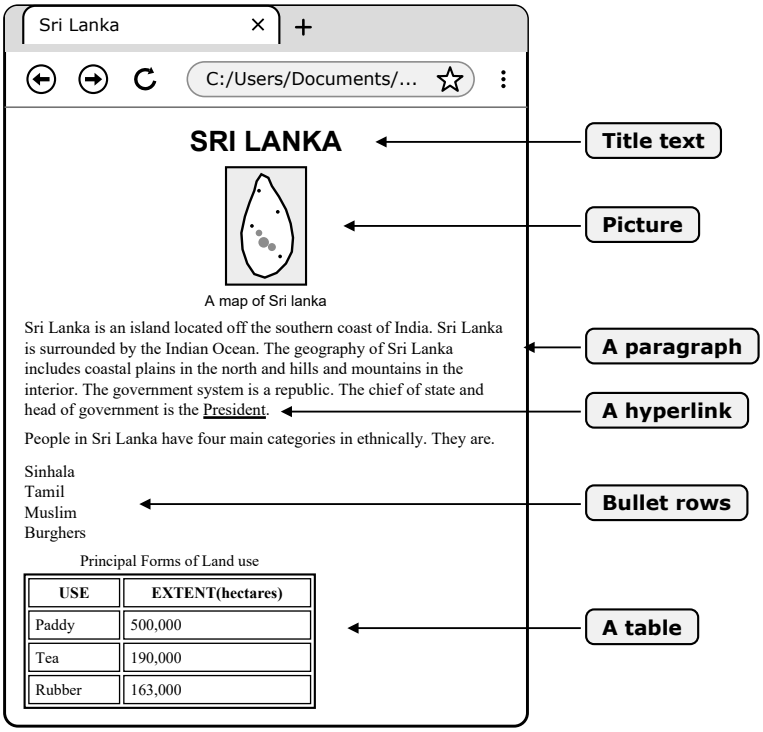


Figure 5.14 - a web page about Sri Lanka

5.12 Inserting an image

Use a low size file type, such as **.png**, **.gif** or **.jpeg**, and keep the images in a separate folder. The image tag **<img>** links an image to the web page. It has no closing tag: it is a **blank (empty) tag**. It has about 12 attributes:

Attribute	The value (Table 5.2)
src	source: the image file name with its extension; if the image is in another folder, give the path.
alt	alternative: the text shown in the image position if the image is not available.
width	the width of the image: pixels or a percentage of the size of the page.
height	the height of the image: pixels or a percentage of the size of the page.
border	the thickness of a border round the picture in pixels; "0" for no border.
align	left, right, top, bottom or middle.

← Example (textbook page 145)

```
<img src = "C:\Users\Pictures\srilanka.jpg"
     alt = "Map" width = "100" height = "200"
     border = "3" align = "middle">
```

WRITING SPACE 7 - The image tag

write what your teacher tells you

5.13 Paragraphs, gaps and line breaks

Paragraphs: ** ** ... ** ** (Figure 5.22). A paragraph is **left aligned** by default; the align attribute can be "left", "right", "center" or "justify" (Table 5.3).

A gap between words: the character entity ** **.

Line breaks: **
** breaks the line exactly where it is. Like **<img>**, it is an empty tag (Figure 5.24).

Example	Tag	Alignment
	** **	Left
	** **	Center
	** **	Right
	** **	Justify

```
<p> The government system is   a republic. The chief of
state and head of government is    the President. </p>          ←  
People in Sri Lanka have four main categories in ethnically.
They are.<br/><br/>
Sinhala<br/>Tamil<br/>Muslim<br/>Burghers                                      ← <br/>
```

5.14 Lists

Three main types of lists (**<li>** gives each item): **(i) unordered** - **<ul>**, its type attribute gives the symbol: **disc**, **square** or **circle**; **(ii) ordered** - **<ol>**, type "1" Hindu Arabic numbers, "A" English capital letters, "a" simple letters, "i" Roman numbers with simple letters, "I" Roman numbers with capital letters (Table 5.4); **(iii) description** - **<dl>** begins it, **<dt>** is the introduction, **<dd>** the description (Table 5.5).

```
<ul type = "disc">
   <li> Sinhala</li>
   <li> Tamil</li>
   <li> Muslim</li>
   <li> Burgher</li>
</ul>
```

- Sinhala
- Tamil
- Muslim
- Burgher

Figure 5.26 - an unordered list

```
<ol type = "I" >
   <li>Sinhala</li>
   <li>Tamil</li>
   <li>Muslim</li>
   <li>Burgher</li>
</ol>
```

I. Sinhala
II. Tamil
III. Muslim
IV. Burgher

Figure 5.27 - an ordered list

```
<dl>
<dt> Main subjects for the science stream </dt>
      <dd> Biology </dd>
      <dd> Physics</dd>
      <dd> Chemistry </dd>
      <dd> Combined Maths </dd>
<dt> Main subjects for the commerce stream </dt>
      <dd> Economics </dd>
      <dd> Business Studies </dd>
      <dd> Accountancy </dd>
</dl>
```

Figure 5.28 - a description list

Main subjects for the science stream
Biology
Physics
Chemistry
Combined Maths
Main subjects for the commerce stream
Economics
Business Studies
Accountancy

- Sinhala
- Tamil
  1. Sri Lankan Tamil
  2. Indian Tamil
- Muslim
- Burgher

Figure 5.29 - a list in a list

5.15 Hyperlinks

The **anchor tag** **<a>** creates a hyperlink for a text or an image. Its most important attribute, **href**, gives the place to link: for a file in the same folder, its name with the file extension; for another web page, the relevant path, accurately (Figure 5.30). Move the mouse over the link to open it.

```
<a href="President.jpeg">President </a>
<a href="President.html">President </a>
<a href="http://www.president.gov.lk/president">President </a>
<a href="http://www.president.gov.lk" >President.</a></p>          ← Figure 5.30
```

5.16 Tables

```
<table border="1" >
  <caption> Principal Forms of Land use</caption>
  <tr>
     <th>USE</th>
     <th>EXTENT(hectares)</th>   </tr>
  <tr>
     <td>Paddy</td>
     <td>500,000</td>   </tr>
  <tr>
     <td>Tea</td>
     <td>190,000</td>   </tr>
  <tr>
     <td>Rubber</td>
     <td>163,000</td>   </tr>
</table>
```

Figure 5.32 - the code of the table

Tag	What it does
<table>	begins a table; its attributes decorate the table
<caption>	a title or a description above the table
<tr>	adds a new row to the table
<td>	a cell with data in a row
<th>	the table header: a cell whose text is bold

Principal Forms of Land use

USE	EXTENT(hectares)
Paddy	500,000
Tea	190,000
Rubber	163,000

Figure 5.33 - the output

Attributes of <table> (Table 5.6): align, width, height, bgcolor, border, bordercolor, background, **cellpadding** (the gap between the cell border and its content) and **cellspacing** (the gap between the cells). **<tr>** and **<td>** take attributes too.

5.17 Merging cells: colspan and rowspan

colspan creates a cell by merging several columns, **rowspan** by merging several rows; the value is the number merged. **<th>**, the table header tag, can be used instead of **<td>** for any cell: its text is bold, and it takes all the attributes of **<td>**.

```
<table border="1">
<tr> <th colspan="2"> <center> LAND USE </center> </th> </tr>
<tr>   <td> USE </td>
       <td> EXTENT(hectares) </td>   </tr>
<tr>   <td> Tea </td>
       <td rowspan="2"> 353,000 </td>   </tr>
<tr>   <td> Rubber </td>
</table>
```

Figure 5.36 - colspan in the first row, rowspan for 353,000

LAND USE	
USE	EXTENT(hectares)
Tea	353,000
Rubber	

WRITING SPACE 8 - Table tags — write what your teacher tells you

WRITING SPACE 9 - Three kinds of lists — write what your teacher tells you

DO IT - PRACTICAL 4 (ACTIVITIES, PAGES 145 - 152) - THE SRI LANKA PAGE

1	Save Figure 5.15 as **srilanka.html**; add the map with **<img>** (Figure 5.21).
2	Add the paragraph (Figure 5.22). Try the align values, ** ** and other character entities.
3	Add the four ethnic groups with ** ** (Figure 5.24), then as lists - also a list in a list.
4	Make **President** a hyperlink (Figure 5.30); link the picture to a website too.
5	List the attributes of **<a>**. Add the table (Figure 5.32); compare with Figure 5.14.

DO IT - PRACTICAL 5 (TEXTBOOK ACTIVITY, PAGE 156) - TABLES

1	Create a web page with this table. Use **rowspan** and **colspan**.
2	Change border, bgcolor, cellpadding and cellspacing; watch the table change.
3	Create the website your team planned (Practical 1) with a suitable HTML editor.

<table>
<tr><th>Column 1</th><th>Column 2</th><th>Column 3</th></tr>
<tr><td rowspan="2">Row 1 Cell 1</td><td>Row 1 Cell 2</td><td>Row 1 Cell 3</td></tr>
<tr><td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
<tr><td colspan="3">Row 3 Cell 1</td></tr>
</table>

A Match the image attributes

6 marks

Write the correct letter in the box.

A - Attribute		B - What it gives
1. **src**		**A.** The thickness of a line round the picture, in pixels.
2. **alt**		**B.** The file name of the image with its extension (or its path).
3. **width**		**C.** Left, right, top, bottom or middle.
4. **height**		**D.** How tall the image is: pixels or a percentage of the page.
5. **border**		**E.** The text shown in the image position when the image is not available.
6. **align**		**F.** How wide the image is: pixels or a percentage of the page.

B Write the output

6 marks

Write in each box what the browser shows (2 marks each).

(a)

```
<ol type = "A">
  <li>Paddy</li>
  <li>Tea</li>
  <li>Rubber</li>
</ol>
```

(b)

```
<ol type = "i">
  <li>Sinhala</li>
  <li>Tamil</li>
  <li>English</li>
</ol>
```

(c)

```
<ul type = "square">
  <li>Kandy</li>
  <li>Galle</li>
  <li>Jaffna</li>
</ul>
```

C Complete the table code

8 marks

Write the missing tag, attribute or value on each numbered line.

land.html - Notepad

```
<❶_____ border="❷_____ ">
<tr> <th ❸_____ ="2"> <center> LAND USE </center> </th> </tr>
<❹_____ >   <td> USE </td>
       <❺_____ > EXTENT(hectares) </td>   </tr>
<tr>   <td> Tea </td>
       <td ❻_____ ="❼_____ "> 353,000 </td>   </tr>
<tr>   <td> Rubber </td>
</❽_____ >
```

LAND USE	
USE	EXTENT(hectares)
Tea	353,000
Rubber	

The output

D Choose the correct answer

8 marks

Underline your answer.

1. Which of these is a low size file type for an image on a web page?
   (a) .docx (b) .html (c) .png (d) .mp4
2. Which tag has no closing tag?
   (a) **<p>** (b) **
** (c) **<b>** (d) **<ol>**
3. A paragraph made with **<p>** and no align value is aligned to the
   (a) centre (b) right (c) left (d) justify
4. ** ** is used to
   (a) start a new line (b) keep a gap between words
   (c) make the text bold (d) insert a picture
5. Which value of the type attribute numbers a list I, II, III, IV?
   (a) "1" (b) "i" (c) "A" (d) "I"
6. In a description list, the description is given with
   (a) **<dl>** (b) **<dt>** (c) **<dd>** (d) **<li>**
7. Which attribute of the anchor tag gives the place to link?
   (a) src (b) alt (c) href (d) align
8. cellspacing is the gap
   (a) between the cells (b) inside a cell
   (c) round the caption (d) under the table

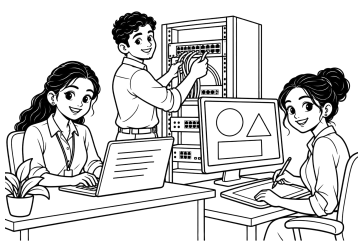
CHAPTER 4

Software, Publishing and Maintenance

? FROM OUR COMPUTER TO THE WORLD

"Our website works on our computer. How can the whole world see it?"

Web designing software, CMS, publishing and maintaining a website (textbook pages 157 - 167).



5.18 Web designing software

HTML coding takes more time and the codes must be changed for updates. Web designing software is quicker: its graphical user interface helps with designing, editing, preview and upload. By the concept **WYSIWYG** (**W**hat **Y**ou **S**ee **I**s **W**hat **Y**ou **G**et) you create web pages the way you design them on the screen. Two types: **web authoring tools** and **Content Management Systems (CMS)**. Web authoring tools prepare the HTML document automatically while you design (5.2.1). Open source: **KompoZer, CKEditor, BlueGriffon**; to be purchased: **Adobe Dreamweaver, Microsoft FrontPage, Visual Studio**.

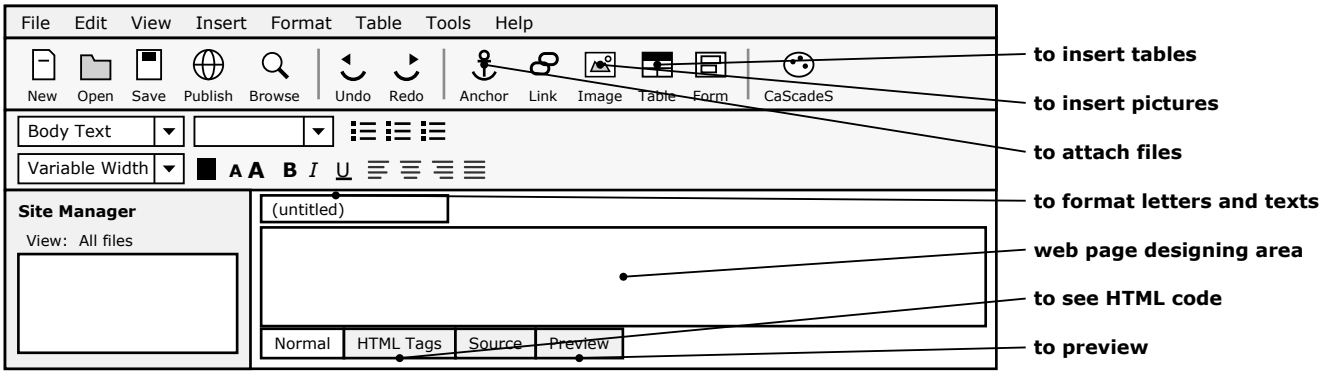


Figure 5.39 - the KompoZer window

DO IT - PRACTICAL 6 (STEPS, PAGES 158 - 160) - KOMPOZER

Six pages on **Systems of the Human Body**: a home page and a page for each of five systems.

1	Install and open KompoZer (www.kompozer.net). Type and format the title **Systems of the Human Body**.
2	Save it (File menu); preview it in a web browser.
3	Insert a table of **6 rows and 3 columns** (Insert menu or Table tool); merge cells as in textbook Figure 5.41.
4	Add texts, images (all in one folder) and hyperlinks.
5	Make a page for each system (**New** tool) in the same folder; link them; preview.

5.19 Static and dynamic websites

By the nature of the content, a website is **static** or **dynamic** (5.2.2). Dynamic websites suit commercial, social or exam holding websites that change often.

	Static website	Dynamic website
Content	not changed for a period of time	often changed
Made with	HTML codes	PHP, ASP.net, JSP; a database or a program
Users	not directly connected to the web server	directly connected to the web server
Security	safer, but coding takes more time	threatened more; needs higher knowledge
Examples	a simple collection of web pages	the current time; exam results by index number

WRITING SPACE 10 - Static and dynamic websites write what your teacher tells you

5.20 Content Management Systems (CMS)

A **CMS** is software for dynamic website designing, content management and publishing; the website can be re-published often with new content. It is very easy and fast, needs no prior knowledge of computer languages and is the most popular tool. With templates and wizards creativity is limited and the websites look monotonous - unless you change the codes (HTML, PHP, JavaScript, CSS, SQL). Open source CMS: **Joomla, WordPress, Drupal** (5.2.3).

DO IT - PRACTICAL 7 (TEXTBOOK ACTIVITY, PAGE 162) - JOOMLA

1	Install **WAMP** or **XAMPP** (one package with Apache, PHP and MySQL) with your teacher.
2	Install Joomla (www.joomla.org; help at docs.joomla.org). Select templates, articles and components.
3	Build the website your team planned; see the preview and make the necessary changes.

5.21 Roles and publishing

Role	The tasks (5.2.4)
Author	Designs the interfaces of the website.
Editor	Writes the codes, checks the errors and rectifies them.
Publisher	Fulfils the tasks of publishing the website.
Administrator	Maintains the web server and the database.
Users	Use the website.

Publication of a website includes designing, hosting, updating and posting. A web publisher needs (5.3):

1. A web designing software
2. A web host
3. A domain name
4. File transfer protocol (FTP)
5. An Internet connection

A **web host** is a computer that keeps the website so that anyone can access it at any time from any place through the Internet. A big institute has its own **dedicated host**; a small one gets space on a **shared host** (a basic fee and an annual charge by the capacity). Free web hosts show their advertisements on your website (5.3.1).

WRITING SPACE 11 - Publishing a website write what your teacher tells you

5.22 Domain names, FTP and the Internet

As a national identity number identifies a person, a **domain name** finds a website. **ICANN** (Internet Corporation for Assigned Names and Numbers) assigns and administers domain names. Register at ICANN or www.godaddy.com, www.bluehost.com, www.nic.lk (a basic fee, renewed every year) - better by the publisher himself (5.3.2).

1. Show the institute or the objective: **nie** in www.nie.lk shows the institute.
2. A suitable top level domain: .com, .edu, .org, .lk (.org in www.wikipedia.org: non commercial).
3. Short: easy to remember and to type accurately.
4. No numbers or lines: **www.myweb2.org** may be heard as **www.mywebtu.org**.

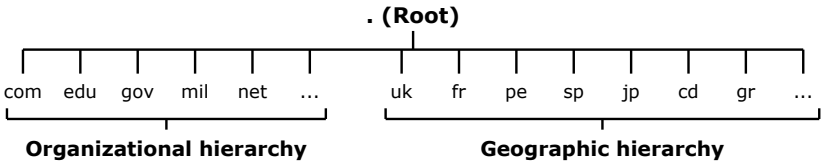


Figure 5.43 - domain names

FTP (File Transfer Protocol) software transfers the website to the web host: Filezilla and SmartFTP; most web tools and shared hosts transfer files too. An **Internet connection** is needed to transfer, update and declare the path; choose the Internet service provider by speed, charges and maintenance: Sri Lanka Telecom, Lanka Bell, Mobitel, Dialog Axiata (5.3.3 - 5.3.4).

5.23 Maintaining a website

1. Update the content daily, weekly or monthly - and also the way it is presented: pictures, colours, shapes. A column on the Home Page for new information shows changes at once.
2. Viruses and unauthorized access can make the website malfunction: pay regular attention, take back ups from time to time and follow proper security methods.
3. When new versions of the software used to design the website come, update the website. It is important for its security.

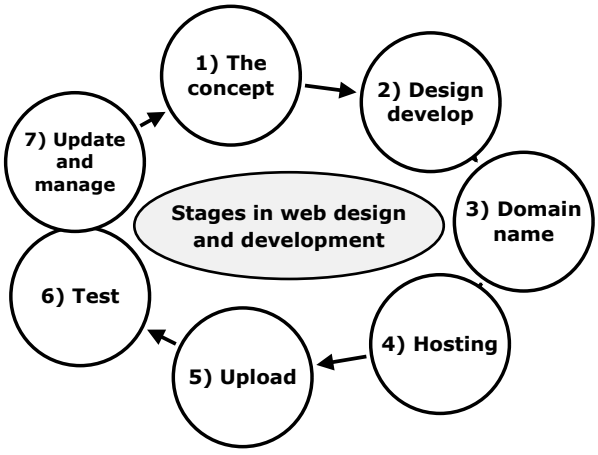


Figure 5.44 - the stages of web design and development

DO IT - PRACTICAL 8 (ACTIVITIES, PAGES 164 - 166) - PUBLISH YOUR WEBSITE

1	Name two institutes that give free web host services, with their web addresses.
2	Visit www.nic.lk and study the steps to obtain a .lk domain name.
3	Publish the website your team designed: a web host, a domain name, FTP. Test every page and link.

A Static or dynamic?

8 marks

Tick the kind of website each sentence is about.

The website ...	Static	Dynamic
Its content is not changed for a period of time.	☐	☐
It is made with PHP, ASP.net or JSP.	☐	☐
Users are not directly connected to the web server.	☐	☐
Its server makes the pages from a database.	☐	☐
It is safer, but the coding takes more time.	☐	☐
It shows exam results for an index number.	☐	☐
Its content changes only when its code is changed.	☐	☐
It suits commercial, social and exam websites.	☐	☐

B Good domain names?

6 marks

Tick one box for each domain name.

Domain name	Good	Not good
www.nie.lk (National Institute of Education)	☐	☐
www.myweb2.org	☐	☐
www.wikipedia.org (a non commercial website)	☐	☐
www.the-best-online-shop-in-sri-lanka.com	☐	☐
www.galleteashop.lk (a tea shop in Galle)	☐	☐
www.abc123.lk (the website of a school)	☐	☐

C Choose the correct answer

8 marks

Underline your answer.

1. Web pages are created in the same way they are designed on the screen. This is
   (a) FTP (b) WYSIWYG (c) ICANN (d) HTML
2. Which of these is an open source web authoring tool?
   (a) Dreamweaver (b) FrontPage (c) Visual Studio (d) KompoZer
3. Which of these is an open source CMS?
   (a) FileZilla (b) Drupal (c) XAMPP (d) GoDaddy
4. Which of these is a part of the maintenance of a website?
   (a) taking back ups (b) a site map (c) the user groups (d) a domain name
5. The institute that assigns and administers domain names is
   (a) ICANN (b) NIE (c) ISP (d) CMS
6. Software that transfers a website to the web host is
   (a) a CMS (b) a browser (c) Notepad (d) FTP software
7. A small institute usually gets space for its website on a
   (a) dedicated host (b) shared host (c) search engine (d) database
8. Who maintains the web server and the database of a website?
   (a) author (b) editor (c) user (d) administrator

REVISION

Unit 5 in One Page

Read this the night before the test

Topic	The key facts
Web applications	Information communication, e Commerce, entertainment, advertising, news, social media, search engine, personal.
Users, content	Users: language and culture, age, education, gender, preferences. Content: all the audio-visual information and services.
Content management	Site map; main and sub menus; brief information; lists; easy navigation; copyright and updated time.
Media resources	Small files: crop, compress; frame size, resolution, frame rate; bit rate, sample rate, channels. Ask permission.
HTML	Hyper Text Markup Language. **<html>** holds **<head>** (**<title>**) and **<body>**. Save as .html, All Files.
Tags	Element, attribute, value: **<font size="5">**. Not case sensitive; **/** closes; no error messages.
Title text	**<h1>** - **<h6>**; **<font>** size 1 - 7 (default 3), face, color (16 names or #RRGGBB); ** **.
Formatting	**<b> <i> <u> <s> <sup> <sub> <marquee>**
Image, text	**<img>** (empty): src alt width height border align; **<p align>**; ** **; ** **.
Lists	** ** disc, square, circle; ** ** 1, A, a, i, I; **<dl> <dt> <dd>**; ** ** for each item.
Hyperlinks	**<a href="...">**: a file name and extension, or the full path of another web page.
Tables	**<table> <caption> <tr> <th> <td>**; colspan, rowspan, cellpadding, cellspacing.
Software	WYSIWYG tools (KompoZer); CMS (Joomla). Static: HTML, safer. Dynamic: PHP, ASP.net, JSP, a database.
Publish, maintain	Software, web host, domain name, FTP, Internet. Update, back up, security, new versions.

TICK WHEN YOU CAN DO IT

- [ ] I can name the kinds of web applications and plan a website for its users.
- [ ] I can organise the content of a website with a site map and menus.
- [ ] I can choose small media files and respect copyright.
- [ ] I can write an HTML document and save and open it as a web page.
- [ ] I can format text and colours and add images, lists, links and tables.
- [ ] I can compare authoring tools and CMS, static and dynamic websites.
- [ ] I can list what is needed to publish and maintain a website.

WORD POWER

Unit 5 Word List

Twenty-six words you must know

Word	What it means
Web application	A kind of website: e Commerce, news, social media ...
User group	The people who access and use a website.
Web content	All the audio-visual information and services of a website.
Content management	The systematic organisation of the selected content of a website.
Site map	A plan of the web pages of a website and the hyperlinks among them.
Main menu / Sub menu	The list of the main parts of a website; a sub menu opens under one item.
Navigation	Moving from page to page with hyperlinks.
Copyright	The owner's right to his work: get permission before you use it.
Fair use	Using media without permission for news reporting, academic purposes or criticism.
HTML	Hyper Text Markup Language: the language of the documents in the World Wide Web.
Tag	An HTML code in angular brackets, e.g. **<p>**.
Element / attribute / value	The three parts of a tag: **<font color="blue">**.
Empty tag	A tag with no closing tag, e.g. **<img>** and ** **.
Hyperlink	A text or an image that opens another file or web page: **<a href>**.
Markup language	A language whose codes show how the content is displayed, e.g. HTML.
Character entity	A code such as ** ** (a gap between words).
colspan / rowspan	Attributes that merge several columns / several rows into one cell.
WYSIWYG	What You See Is What You Get: design pages as they will look.
Web authoring tool	Software that makes the HTML while you design, e.g. KompoZer.
CMS	Content Management System: designs, manages and publishes dynamic websites.
Static website	A website whose content is not changed for a period of time.
Dynamic website	A website whose content is often changed, made from a database.
Web host	A computer that keeps a website for anyone to reach at any time.
Domain name	The registered name of a website, e.g. nie in www.nie.lk.
ICANN	Internet Corporation for Assigned Names and Numbers.
FTP	File Transfer Protocol: software that transfers a website to the web host.

"Learn the words today, and you will speak like a web designer tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 5 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. www.amazon.com is an example of which kind of web application?
   (a) News (b) e Commerce (c) Personal (d) Search engine
2. Which of these helps us to understand the users of a website?
   (a) their age (b) the frame rate (c) the bit rate (d) the pixels
3. Which of these reduces the size of a graphic?
   (a) sample rate (b) Crop / Trim (c) channels (d) frame rate
4. In HTML, the letter M stands for
   (a) Making (b) Markup (c) Media (d) Menu
5. The two main parts of an HTML document are the head and the
   (a) title (b) table (c) body (d) font
6. The default font size of the font tag is
   (a) 1 (b) 3 (c) 5 (d) 7
7. Software for dynamic website designing, content management and publishing is
   (a) a CMS (b) Notepad (c) FTP (d) a browser
8. The last stage of web design and development is
   (a) Upload (b) Hosting
   (c) Test (d) Update and manage

2 Fill in the blanks

8 marks

One word in each blank.

1. A site ..... shows the web pages and the hyperlinks among them.
2. Using media of others without permission may lead to ..... issues.
3. HTML tags are written inside ..... brackets.
4. The ..... attribute of the font tag gives the type of letters.
5. The ..... tag aligns a title or a picture to the centre.
6. The **
** tag gives a line ..... .
7. Websites based on HTML language are ..... websites.
8. nie in www.nie.lk is part of the ..... name.

3 Which tag made it?

8 marks

Write the HTML tag that made each output.

1. SRI LANKA

2. ~~SRI LANKA~~

3. 10^2

4. H_2O

5. • Sinhala
   • Tamil
   • Muslim

6. I. Sinhala
   II. Tamil
   III. Muslim

7. 

8. President

4 Answer these

16 marks

Write in the space given.

1. Name two kinds of web applications and give one example website of each. (2 marks)

2. Write two ways to reduce the size of a video file for a website. (2 marks)

3. Write the HTML code to show the text SRI LANKA in red, in size 5, with the face Arial. (3 marks)

4. Write the HTML code for a list that shows A. Tea B. Rubber (3 marks)

5. Write two differences between a static website and a dynamic website. (2 marks)

6. Write four things a web publisher needs to publish a website. (2 marks)

7. Write two things to consider when you select a domain name. (2 marks)